

MATERIAL SAFETY DATA SHEET

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

MANUFACTURER:

WESMAR PRODUCTS, INC.
1440 N.W. 45th STREET
SEATTLE, WA 98107
(206) 782-8188

INFORMATION PHONE:**EMERGENCY PHONE:****PRODUCT NAME:** NC BRAKE CLEAN**PRODUCT NUMBER:****PREPARED BY:** Patricia Rodabaugh**DATE PREPARED:** 04/13/2001**SYNONYMS:**

WM 5805

WM 5805X

WM 5806

2. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Name	CAS	OSHA PEL	ACGIH TLV	Weight %
Ethyl alcohol	64-17-5	1000 ppm	1000 ppm	1.5 - 3
Isopropyl alcohol	67-63-0	400 ppm	400 ppm	<1.0
Methyl isobutyl ketone	108-10-1	50 ppm	50 ppm	<1
Hexane, n-	110-54-3	50 ppm	50 ppm	89-94
Methyl alcohol	67-56-1	200 ppm	200 ppm	4-7

3. HAZARDOUS IDENTIFICATION

EMERGENCY OVERVIEW:

DANGER! Poison. Extremely Flammable. Vapor harmful. May be fatal or cause blindness if swallowed. Avoid prolonged breathing of vapors. Avoid contact with eyes and skin.

POTENTIAL HEALTH EFFECTS

EYE CONTACT: Material may cause eye irritation. Direct contact with the liquid or exposure to its vapors or mists may cause stinging, tearing, redness, swelling and eye damage.

INHALATION: Excessive exposure to this product may cause headache, CNS depression, drowsiness, dizziness, loss of appetite, irritation of the respiratory tract, drunkenness, unconsciousness, or death.

INGESTION: POISONOUS. May be fatal or cause blindness if swallowed. Ingestion may have a narcotic effect including signs of CNS depression such as dizziness, headache, drowsiness, loss of coordination, and fatigue.

SKIN CONTACT: Liquid is moderately irritating to the skin. Prolonged or repeated contact can result in defatting and drying of the skin which may result in skin irritation and dermatitis (rash).

SIGNS AND SYMPTOMS OF EXPOSURE:

Shortness of breathing, confused behavior, redness of skin, swelling of tissues, watery eyes, and nausea.

4. FIRST AID MEASURES

EYE CONTACT: Immediately flush eyes with plenty of water for at least 15 minutes while holding eyelids open. Get medical attention.

INHALATION: Remove victim to fresh air. If not breathing, give artificial respiration. If breathing is difficult, oxygen should be administered by qualified personnel. Get medical attention.

INGESTION: Do not give liquids if victim is unconscious or drowsy. Otherwise, give 2 glasses of water and induce vomiting by giving 30cc syrup of ipecac (or touching finger to the back of victim's throat). Keep victim's head below hips while vomiting. Call doctor.

SKIN CONTACT: Remove contaminated clothing/shoes. Flush skin with water. Follow by washing with soap and water. If irritation occurs, get medical attention. Do not reuse clothing until cleaned.

AGGRAVATED MEDICAL CONDITIONS:

Preexisting eye, skin and respiratory disorders may be aggravated by exposure to this product. Impaired function from preexisting disorders may be aggravated by exposure to this product.

SUPPLEMENTAL HEALTH INFORMATION:

See last page of MSDS for Supplemental Health Information.

5. FIRE FIGHTING MEASURES

FLAMMABLE PROPERTIES

FLASH POINT: <10 F

FLASH POINT METHOD USED: Tag Closed Cup

AUTOIGNITION: N/A

FLAMMABLE LIMITS: LEL: 1.10% UEL: 36.00%

EXTINGUISHING MEDIA:

Use water fog, "alcohol" foam, dry chemical, or CO2.

SPECIAL FIRE-FIGHTING PROCEDURES:

WARNING. Flammable Liquid. Clear fire area of unprotected personnel. Do not enter confined fire space without full bunker gear, including a positive pressure NIOSH approved SCBA. Cool fire exposed containers with water.

UNUSUAL FIRE AND EXPLOSION HAZARDS:

When heated above the flash point this material emits flammable vapors which, when mixed with air, can burn or be explosive. Fine mist or spray may be flammable at temperatures below the flash point.

COMBUSTION PRODUCTS:

Carbon monoxide and unidentified organic compounds may be formed during combustion.

6. ACCIDENTAL RELEASE MEASURES

STEPS TO BE TAKEN IN CASE MATERIAL IS SPILLED OR RELEASED

WARNING. Flammable. Ventilate area of leak or spill. Remove all sources of ignition. Clean-up personnel require protective clothing and respiratory protection from vapors. Only specially trained or qualified personnel should handle the emergency.

7. HANDLING AND STORAGE

PRECAUTIONS TO BE TAKEN IN HANDLING AND STORING

Keep away from heat, sparks, and flame. Surfaces that are hot may ignite even liquid product in the absence of sparks or flame. Extinguish pilot lights, cigarettes and turn off other sources of ignition prior to use and until all vapors are gone.

OTHER PRECAUTIONS:

KEEP OUT OF REACH OF CHILDREN! Empty containers retain product residue and can be dangerous. Do not pressurize, cut weld, braze, solder, drill, grind, or expose such containers to heat, flame, sparks static electricity, or other sources of ignition.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

PERSONAL PROTECTIVE EQUIPMENT

RESPIRATORY PROTECTION (Specify Type):

If exposure may or does exceed occupational exposure limits (Sec. 2) use a NIOSH approved respirator to prevent overexposure. In accord with 29 CFR 1910.134 use either an atmosphere-supplying respir. of an air-purifying respir. for organic vapors.

VENTILATION:

Provide exhaust ventilation sufficient to keep the airborne concentration of this product below its exposure limits. Exhaust air may need to be cleaned by scrubbers or filters to reduce environmental contamination.

PROTECTIVE GLOVES:

Test data indicate the best protection is provided by neoprene, nitrile, and natural rubber gloves.

EYE PROTECTION:

Use chemical safety goggles and/or full face shield where splashing is possible. Contact lenses should not be worn when working with this material. Maintain eye wash fountain and quick-drench facilities in work areas.

OTHER PROTECTIVE CLOTHING OR EQUIPMENT

Where splashing is possible, full chemically resistant protective clothing (e.g., acid suit) and boots are required.

WORK/HYGIENIC PRACTICES:

Use good personal hygiene when handling this product. Wash hands after use, before eating, drinking, smoking, or using the toilet.

ENGINEERING CONTROLS:

Facilities storing or utilizing this material should be equipped with and eyewash facility and a safety shower.

EXPOSURE GUIDELINES

May be harmful or fatal if swallowed. May irritate body tissues. Use with adequate ventilation. Avoid breathing vapor. Do not get in eyes, on skin, on clothing. Wash thoroughly after handling.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE AND ODOR:

Colorless, clear liquid. Mild odor.

SOLUBILITY IN WATER:

Infinite in water

BOILING POINT: 147 F

PERCENT VOLATILE: 100

VAPOR PRESSURE: 70.9 - 124 mmHg. @ 20 C

PH: essentially neutral

SPECIFIC GRAVITY: 0.63

MOLECULAR WEIGHT: N/A

MELTING POINT: NDA

POUNDS PER GALLON: 5.28

FREEZING POINT: NDA

OTHER PROPERTIES:

EVAPORATION RATE: Slower than ether

VAPOR DENSITY: Heavier than air

10. STABILITY AND REACTIVITY

STABILITY: Stable

CONDITIONS TO AVOID:

Avoid heat, sparks, flame and contact with strong oxidizing agents. Do not store or handle in aluminum equipment at temperatures above 120 deg. F.

INCOMPATIBILITY:

Strong acids, bases, oxidizers, alkali metals, and halogenes.

HAZARDOUS DECOMPOSITION OR BY PRODUCTS

Carbon monoxide and unidentified organic compounds may be formed during combustion.

HAZARDOUS POLYMERIZATION: Will Not Occur

CONDITIONS TO AVOID:

Avoid heat, flame, and other sources of ignition. Do not store or handle in aluminum equipment at temperatures above 120 F.

11. TOXICOLOGICAL INFORMATION

Poison. May be fatal or cause blindness, if swallowed. Cannot be made nonpoisonous.

12. ECOLOGICAL INFORMATION

Avoid uncontrolled releases of this material. Where spills are possible, a comprehensive spill response plan should be developed and implemented.

13. DISPOSAL CONSIDERATIONS

The preferred options for disposal are to send to licensed reclaimers, or to permitted incinerators. Any disposal practice must be in compliance with federal, state, and local regulations. Do not dump into sewers, ground, or any body of water.

14. TRANSPORT INFORMATION

DOT CLASS: Flammable liquid

HAZARD CLASS: 3

UN NUMBER: UN 1992

PACKING GROUP: II

GUIDE NUMBER: 131

PROPER SHIPPING NAME: Flammable liquids, toxic, n.o.s. (hexane, methanol)

15. REGULATORY INFORMATION

This product is listed on the EPA/TSCA inventory of chemical substances. Per 40 CFR part 82, this product does not contain nor was it directly manufactured with any class I or class II ozone depleting substance.

16. OTHER INFORMATION

HMIS INFORMATION: HEALTH: 3 FLAMMABILITY: 3 REACTIVE: 0 PROTECTION: H

SARA Title III

For SARA Sec. 311/312 reporting, this product should be reported as an immediate (acute) health hazard, and a fire hazard. To the best of our knowledge, none of the chemicals in this product are found on the extremely hazardous substance list (Sec. 302). Methyl isobutyl ketone and methanol are listed as a toxic chemical on the SARA Title III, Section 313 list.

Supplemental Health Info.

In response to a TSCA test rule, several studies of isopropanol (IPA) have been completed. The studies and their results are as follows: 1) Both mutagenicity studies, the mouse mononucleus and CHD assays, were negative. 2) Rat and rabbit oral teratogenicity and developmental toxicology: A) There was no evidence that IPA caused teratogenicity in rats or rabbits. B) Developmental toxicity was seen in rats at 1200 mg/kg (evidenced by decreased body weight**) while no developmental toxicity was seen in the rabbit study. For rats, the NOAEL was 400 mg/kg; for rabbits, 480 mg/kg. This work also identified pregnant rabbits to be approximately eight times more sensitive to IPA's lethal effects than non-pregnant rabbits**. In response to a TSCA test rule, several studies of isopropanol (IPA) have been completed. The studies and their results are as follows: 1) Both mutagenicity studies, the mouse mononucleus and CHD assays, were negative. 2) Rat and rabbit oral teratogenicity and developmental toxicology: A) There was no evidence that IPA caused teratogenicity in rats or rabbits. B) Developmental toxicity was seen in rats at 1200 mg/kg (evidenced by decreased body weight**) while no developmental toxicity was seen in the rabbit study. For rats, the NOAEL was 400 mg/kg; for rabbits, 480 mg/kg. This work also identified pregnant rabbits to be approximately eight times more sensitive to IPA's lethal effects than non-pregnant rabbits**.

Other test rule related studies: 3) Rat oral reproductive toxicity: In the rat reproductive toxicity study, the NOAEL for reproduction indices was 1000 mg/kg, however, there was a marked increase in post-weaning pup mortality at this level**; The NOAEL for this effect was 500 mg/kg. Also, the NOAEL for parental female body weight decrease was 100 mg/kg. 4) In rat inhalation neurotoxicity and oral developmental neurotoxicity studies, there was no evidence that IPA caused neurotoxicity in adults (max dose 5000 ppm) or offspring (max dose 1200 mg/kg). 5) Subchronic rat and mouse inhalation toxicity: The subchronic NOAEL was 500 ppm based on clinical signs of CNS depression (both species) and increased body weight and blood effects (rat only) seen at 1500 ppm. (note: The information tagged by "*" above were submitted to EPA under the requirements of TSCA: 8(e).)

OTHER ADDITIONAL INFORMATION:

NA= Not applicable; NDA= No data available

The information contained herein is based on the data available to us and is believed to be accurate. However,
I makes no warranty, expressed or implied regarding the accuracy of these data or the results to be
obtained from the use thereof. assumes no responsibility for injuries from the use of the product described
herein.